

SMART MATERIALS IN CIVIL ENGINEERING

ABSTRACT

Smart materials are an emerging class of materials that have the ability to sense environmental changes and respond to them in a predictable and controlled manner. In the field of civil engineering, smart materials are increasingly used to enhance the performance, safety, and durability of structures.

These materials exhibit properties such as shape memory effect, piezoelectric behavior, self-healing capability, and adaptive response to external stimuli like load, temperature, vibration, and electric or magnetic fields. Examples of smart materials used in civil engineering include shape memory alloys, piezoelectric sensors and actuators, fiber-reinforced polymers with embedded sensors, self-healing concrete, and magnetorheological materials.

The application of smart materials enables structural health monitoring, damage detection, vibration control, and improved resistance to seismic and dynamic loads. The use of smart materials contributes to reduced maintenance costs, extended service life of structures, and the development of intelligent and sustainable infrastructure systems.